

# EL — Elements of Life

OCR B (Salters) H433 storyline **EL**. Where the elements come from: atomic structure and spectra, isotopes and mass spectrometry, the periodic table and Group 2, bonding and the shapes of molecules, amount of substance and titrations.

## 1. Atomic Structure

### The Atom

**Atom:** The smallest particle of an element that can take part in a chemical reaction. Atoms consist of a small, dense, positively charged nucleus containing protons and neutrons, surrounded by electrons in energy levels (shells).

| Particle | Relative Mass      | Relative Charge | Location          |
|----------|--------------------|-----------------|-------------------|
| Proton   | 1                  | +1              | Nucleus           |
| Neutron  | 1                  | 0               | Nucleus           |
| Electron | $1/1836 \approx 0$ | -1              | Shells / orbitals |

The **atomic number (Z)** is the number of protons in the nucleus and defines the element. The **mass number (A)** is the total number of protons and neutrons. For a neutral atom the number of electrons equals the number of protons.

### How We Know: Developing Models of the Atom

H433 asks you to describe how the model of the atom was built up as a succession of gradually more sophisticated pictures, each one adopted because an experiment made the previous one untenable. This is a "how science works" question, and the marks are for the *evidence*, not just the names and dates.

| Model                      | What it said                                                                                            | The evidence that forced the next step                                                                               |
|----------------------------|---------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Dalton (1803)              | Elements are made of identical, indivisible solid spheres                                               | Thomson's discovery of the electron in 1897 showed atoms contain smaller charged parts                               |
| Thomson, "plum pudding"    | A sphere of diffuse positive charge with electrons embedded in it                                       | The Geiger–Marsden scattering experiment (see below)                                                                 |
| Rutherford, nuclear (1911) | A tiny, dense, positive nucleus with electrons somewhere outside it                                     | Atomic line spectra, and successive ionisation enthalpies, show electrons are restricted to particular energy levels |
| Bohr (1913)                | Electrons occupy fixed energy levels (shells) at set distances from the nucleus                         | Finer detail in spectra and the filling order of the elements needed sub-shells                                      |
| Modern quantum model       | Electrons occupy orbitals — regions of space with a given shape and energy, holding up to two electrons | —                                                                                                                    |

The decisive experiment was carried out by **Geiger and Marsden** in Rutherford's laboratory. A narrow beam of alpha particles — helium nuclei, positively charged and relatively massive — was fired at a sheet of gold foil only a few thousand atoms thick, and a movable fluorescent screen was used to count the flashes produced by particles arriving at each angle. Three observations mattered:

- **The overwhelming majority passed straight through, undeflected.** So the atom must be mostly empty space.
- **A small number were deflected through appreciable angles.** So somewhere in the atom there is a concentration of positive charge that repels the positive alpha particle.
- **A very few — a tiny fraction of the beam — came back on themselves, deflected through more than 90°.** So that positive charge must be both very concentrated and very massive, because only something far heavier than an alpha particle could turn one round. This is the nucleus.

None of this is compatible with the plum-pudding model, in which the positive charge is spread thinly through the whole atom: a diffuse positive sphere could never produce a large-angle deflection, let alone a reversal.

**Exam Tip:** Answer this style of question as observation → deduction pairs. "Most passed straight through, so the atom is mostly empty space. A very few were deflected backwards, so the positive charge and nearly all the mass are concentrated in a tiny nucleus." An answer that lists the observations without saying what each one proves — or the reverse — will only get half the marks.

## Isotopes

**Isotopes:** Atoms of the same element (same number of protons) with different numbers of neutrons and therefore different mass numbers. Isotopes have identical chemical properties because they have the same electron configuration, but different physical properties (e.g. density, rate of diffusion).

Examples: carbon-12 ( $^{12}\text{C}$ , 6p + 6n) and carbon-13 ( $^{13}\text{C}$ , 6p + 7n). The existence of isotopes means most elements do not have whole-number relative atomic masses.

## Mass Spectrometry

A mass spectrometer measures the relative masses and abundances of isotopes or molecules. The basic steps are:

1. **Ionisation:** The sample is vaporised and bombarded with high-energy electrons to form positive ions (electron impact ionisation) — an electron is knocked off each atom/molecule.
2. **Acceleration:** Ions are accelerated by an electric field so they all have the same kinetic energy.
3. **Deflection:** Ions pass through a magnetic field. Lighter ions and ions with higher charge are deflected more.
4. **Detection:** Ions reach the detector, producing a current proportional to their abundance.

The mass spectrum shows peaks at each  $m/z$  value. The relative atomic mass ( $A_r$ ) of an element is calculated from the mass spectrum using:

$$A_r = \Sigma(\text{isotope mass} \times \% \text{ abundance}) / 100$$

**Exam Tip:** In mass spectra of molecules, the peak with the highest  $m/z$  value (ignoring the  $M+1$  peak) is the molecular ion peak ( $M^+$ ), giving the relative molecular mass. Fragmentation patterns help

identify functional groups and structural features.

### Worked Example — Calculating $A_r$ :

Chlorine has two isotopes:  $^{35}\text{Cl}$  (abundance 75.8%) and  $^{37}\text{Cl}$  (abundance 24.2%).

$$A_r = (35 \times 75.8 + 37 \times 24.2) / 100 = (2653.0 + 895.4) / 100 = \mathbf{35.5}$$

## Electron Configuration

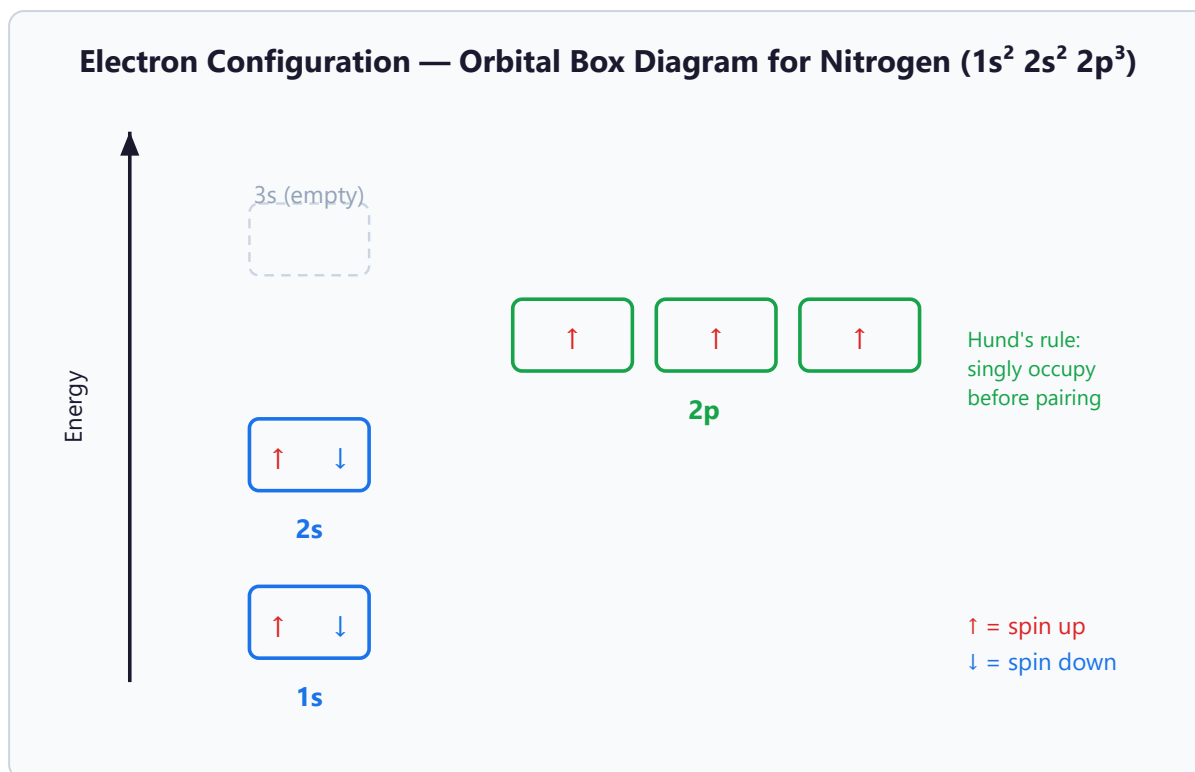
Electrons occupy atomic orbitals. An orbital is a region of space around the nucleus where there is a high probability of finding an electron. Each orbital holds a maximum of two electrons with opposite spins (Pauli exclusion principle).

| Sub-shell | Number of orbitals | Max electrons | Shape                |
|-----------|--------------------|---------------|----------------------|
| s         | 1                  | 2             | Spherical            |
| p         | 3                  | 6             | Dumbbell             |
| d         | 5                  | 10            | Complex / cloverleaf |
| f         | 7                  | 14            | Complex              |

The order of filling follows the Aufbau principle: 1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p, 5s, 4d, 5p ... Note that 4s fills before 3d. When writing electron configurations for ions, remove electrons from the highest principal energy level first (e.g. Fe:  $[\text{Ar}] 3d^6 4s^2$ , but  $\text{Fe}^{2+}$ :  $[\text{Ar}] 3d^6$ ).

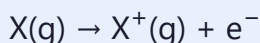
**Hund's rule:** Electrons occupy orbitals of the same sub-shell singly before pairing up, and with parallel spins. This minimises electron-electron repulsion.

Chromium and copper are exceptions: Cr is  $[\text{Ar}] 3d^5 4s^1$  (not  $3d^4 4s^2$ ) and Cu is  $[\text{Ar}] 3d^{10} 4s^1$  (not  $3d^9 4s^2$ ) because half-filled and fully-filled d sub-shells have extra stability due to symmetrical electron distribution and favourable exchange energy.



## Ionisation Energies

**First ionisation energy:** The energy required to remove one mole of electrons from one mole of gaseous atoms to form one mole of gaseous  $1+$  ions.



Ionisation energy depends on: nuclear charge, atomic radius, and shielding (screening) by inner electrons.

### Trends:

- **Across a period:** Generally increases because nuclear charge increases while shielding remains roughly constant, so the outer electron is held more tightly.
- **Down a group:** Decreases because the atomic radius increases and shielding increases, outweighing the increase in nuclear charge.

There are two anomalies across Period 3: a dip from Mg to Al (the 3p electron in Al is easier to remove than a 3s electron in Mg because the 3p sub-shell is higher in energy and slightly shielded by 3s) and a dip from P to S (the paired 3p electron in S experiences extra repulsion, making it easier to remove).

**Exam Tip:** Successive ionisation energies can reveal the number of electrons in each shell. A large jump between successive values indicates that the next electron is being removed from a shell closer to the nucleus (less shielded, more tightly held).

## 2. Amount of Substance

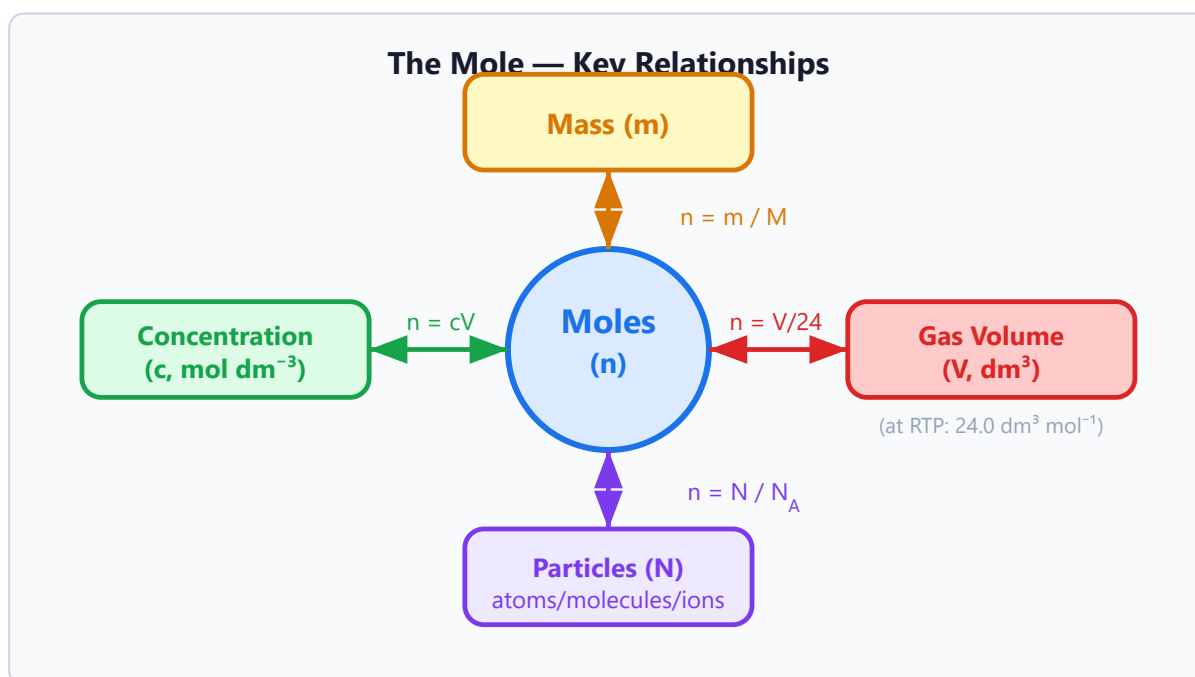
### The Mole and Avogadro's Constant

**Mole:** The amount of substance that contains  $6.022 \times 10^{23}$  particles (atoms, molecules, ions, etc.). This number is Avogadro's constant ( $N_A$ ).

**Molar mass (M):** The mass of one mole of a substance, in  $\text{g mol}^{-1}$ . Numerically equal to the relative formula mass.

$$n = m / M$$

where  $n$  is the number of moles,  $m$  is the mass in grams, and  $M$  is the molar mass in  $\text{g mol}^{-1}$ .



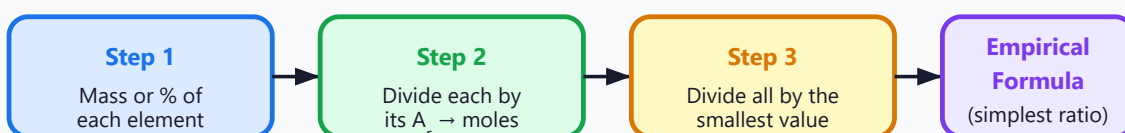
### Empirical and Molecular Formulae

**Empirical formula:** The simplest whole-number ratio of atoms of each element in a compound.

**Molecular formula:** The actual number of atoms of each element in one molecule of a compound.

To find the empirical formula from percentage composition: divide each percentage by the element's  $A_r$ , then divide all results by the smallest value to get the simplest whole-number ratio. The molecular formula is a whole-number multiple of the empirical formula: molecular formula = (empirical formula)  $\times n$ , where  $n = M_r / \text{empirical formula mass}$ .

### Finding Empirical & Molecular Formulae



**Worked Example: Compound with 40.0% C, 6.7% H, 53.3% O**

| Element | % / $A_r$   | Moles | Ratio |
|---------|-------------|-------|-------|
| C       | 40.0 / 12.0 | 3.33  | 1     |
| H       | 6.7 / 1.0   | 6.70  | 2     |
| O       | 53.3 / 16.0 | 3.33  | 1     |

**Empirical formula =  $\text{CH}_2\text{O}$**

If  $M_r = 60$ :  $n = 60/30 = 2$ , so molecular formula =  $\text{C}_2\text{H}_4\text{O}_2$  (ethanoic acid)

## Molar Gas Volume

At room temperature and pressure (RTP, approximately 25 °C and 1 atm), one mole of any gas occupies approximately 24.0 dm<sup>3</sup> (24,000 cm<sup>3</sup>).

$$n = V / V_m$$

where  $V$  is the volume in dm<sup>3</sup> and  $V_m = 24.0 \text{ dm}^3 \text{ mol}^{-1}$  at RTP.

The ideal gas equation provides a more general relationship:

$$pV = nRT$$

where  $p$  is pressure in Pa,  $V$  is volume in  $\text{m}^3$ ,  $n$  is moles,  $R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$ , and  $T$  is temperature in K.

## Concentrations of Solutions

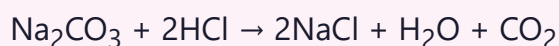
$$c = n / V$$

where  $c$  is concentration in  $\text{mol dm}^{-3}$ ,  $n$  is moles of solute, and  $V$  is volume of solution in  $\text{dm}^3$ . To convert  $\text{cm}^3$  to  $\text{dm}^3$ , divide by 1000.

**Exam Tip:** Always check your units. A very common error is using volume in  $\text{cm}^3$  instead of  $\text{dm}^3$  in the concentration formula. Remember:  $1 \text{ dm}^3 = 1000 \text{ cm}^3$ . For  $pV = nRT$ , pressure must be in Pa, volume in  $\text{m}^3$ , and temperature in K.

### Worked Example — Titration Calculation:

$25.0 \text{ cm}^3$  of  $\text{Na}_2\text{CO}_3$  solution of unknown concentration was titrated with  $0.100 \text{ mol dm}^{-3}$  HCl. The mean titre was  $27.60 \text{ cm}^3$ .



$$n(\text{HCl}) = 0.100 \times 27.60/1000 = 2.76 \times 10^{-3} \text{ mol}$$

$$n(\text{Na}_2\text{CO}_3) = 2.76 \times 10^{-3} / 2 = 1.38 \times 10^{-3} \text{ mol (1:2 ratio)}$$

$$[\text{Na}_2\text{CO}_3] = 1.38 \times 10^{-3} / 0.0250 = \mathbf{0.0552 \text{ mol dm}^{-3}}$$

## Balanced Equations and Reacting Masses

A balanced chemical equation shows the molar ratios of reactants and products. Use these ratios to calculate reacting masses, volumes of gases, and volumes of solutions. The general approach:

1. Write the balanced equation.
2. Calculate moles of the known substance.
3. Use the molar ratio to find moles of the required substance.

4. Convert moles to the desired quantity (mass, volume, concentration).

## Percentage Yield and Atom Economy

$$\text{Percentage yield} = (\text{actual yield} / \text{theoretical yield}) \times 100\%$$

$$\text{Atom economy} = (\text{M}_r \text{ of desired product} / \Sigma \text{M}_r \text{ of all products}) \times 100\%$$

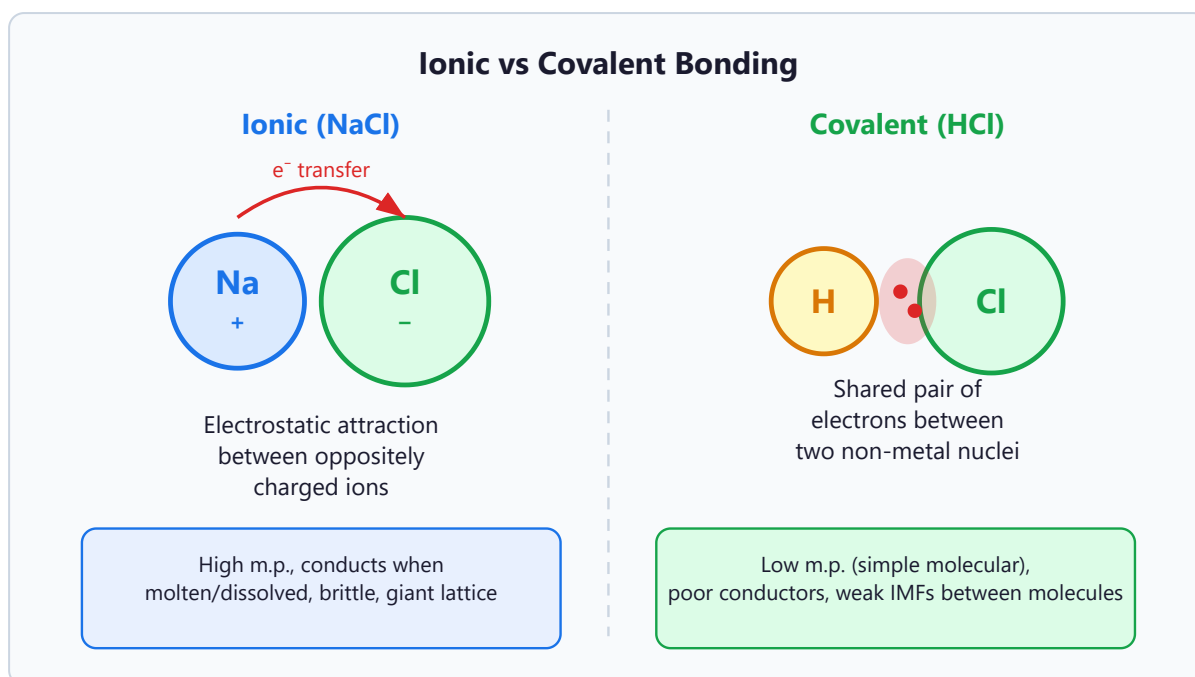
A high atom economy means less waste. Addition reactions have 100% atom economy. Substitution and elimination reactions have lower atom economies because by-products are formed.

## 3. Bonding and Structure

### Ionic Bonding

**Ionic bonding:** The electrostatic attraction between oppositely charged ions, formed by the transfer of electrons from a metal to a non-metal. Metals lose electrons to form cations; non-metals gain electrons to form anions.

Ionic compounds form giant ionic lattices (e.g. NaCl). Properties: high melting points (strong electrostatic forces throughout the lattice), conduct electricity when molten or dissolved (ions are free to move), brittle (displacement of layers causes like charges to align and repel), and dissolve in polar solvents such as water.



## Covalent Bonding

**Covalent bond:** A shared pair of electrons between two atoms, where both nuclei are attracted to the shared electron density. This occurs between non-metal atoms.

**Dative (coordinate) bond:** A covalent bond in which both electrons in the shared pair come from the same atom (the donor). Represented by an arrow from donor to acceptor. Example:  $\text{NH}_4^+$  (the lone pair on nitrogen is donated to  $\text{H}^+$ ).

A **single bond** consists of one shared pair ( $\sigma$  bond). A **double bond** consists of two shared pairs (one  $\sigma$  + one  $\pi$  bond). A **triple bond** consists of three shared pairs (one  $\sigma$  + two  $\pi$  bonds).

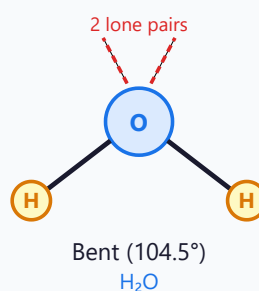
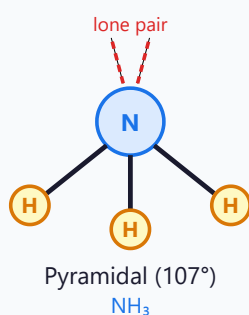
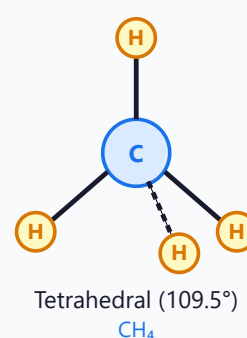
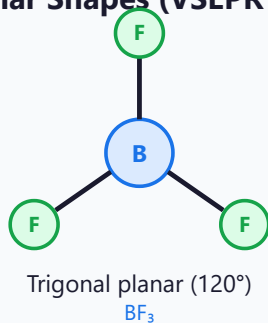
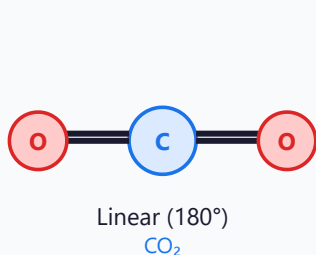
## Shapes of Molecules — VSEPR Theory

**VSEPR (Valence Shell Electron Pair Repulsion) theory:** Electron pairs around a central atom arrange themselves to be as far apart as possible to minimise repulsion. Lone pairs repel more than bonding pairs: lone pair–lone pair > lone pair–bonding pair > bonding pair–bonding pair.

| Electron pairs | Bonding pairs | Lone pairs | Shape | Bond angle | Example |
|----------------|---------------|------------|-------|------------|---------|
|----------------|---------------|------------|-------|------------|---------|

|   |   |   |                      |            |                                                |
|---|---|---|----------------------|------------|------------------------------------------------|
| 2 | 2 | 0 | Linear               | 180°       | CO <sub>2</sub> , BeCl <sub>2</sub>            |
| 3 | 3 | 0 | Trigonal planar      | 120°       | BF <sub>3</sub> , AlCl <sub>3</sub>            |
| 3 | 2 | 1 | Bent / V-shape       | ~117°      | SO <sub>2</sub>                                |
| 4 | 4 | 0 | Tetrahedral          | 109.5°     | CH <sub>4</sub> , NH <sub>4</sub> <sup>+</sup> |
| 4 | 3 | 1 | Trigonal pyramidal   | 107°       | NH <sub>3</sub>                                |
| 4 | 2 | 2 | Bent / V-shape       | 104.5°     | H <sub>2</sub> O                               |
| 5 | 5 | 0 | Trigonal bipyramidal | 90° & 120° | PCl <sub>5</sub>                               |
| 6 | 6 | 0 | Octahedral           | 90°        | SF <sub>6</sub>                                |

### Molecular Shapes (VSEPR Theory)



Lone pairs reduce bond angles: LP-LP > LP-BP > BP-BP repulsion

## Electronegativity and Polarity

**Electronegativity:** The ability of an atom to attract the bonding pair of electrons in a covalent bond towards itself. Measured on the Pauling scale. Fluorine is the most electronegative element (4.0).

**Trends:** Electronegativity increases across a period (greater nuclear charge, smaller atomic radius) and decreases down a group (greater atomic radius, more shielding).

A **polar bond** exists when there is a difference in electronegativity between two bonded atoms. The more electronegative atom carries a partial negative charge ( $\delta^-$ ) and the less electronegative atom carries a partial positive charge ( $\delta^+$ ). A molecule is polar overall only if the individual bond dipoles do not cancel (i.e. the molecule is not symmetrical). For example,  $\text{CO}_2$  has polar bonds but is a non-polar molecule (linear, dipoles cancel), whereas  $\text{H}_2\text{O}$  is a polar molecule (bent shape, dipoles do not cancel).

## Metallic Bonding

**Metallic bonding:** The electrostatic attraction between a lattice of positive metal ions and a sea of delocalised electrons. The outer-shell electrons are free to move throughout the structure.

Properties of metals: high melting points (strong metallic bonds), good electrical conductors (delocalised electrons carry charge), good thermal conductors (delocalised electrons transfer kinetic energy), malleable and ductile (layers of ions can slide over each other without breaking the bonding).

## Intermolecular Forces

Intermolecular forces are weak forces between molecules. They determine physical properties such as boiling point and solubility.

| Type                       | Strength | Origin                                                                                                      | Present in      |
|----------------------------|----------|-------------------------------------------------------------------------------------------------------------|-----------------|
| London (dispersion) forces | Weakest  | Temporary instantaneous dipoles caused by random electron movement induce dipoles in neighbouring molecules | All molecules   |
| Permanent dipole–dipole    | Moderate | Attraction between the $\delta^+$ end of one polar molecule and the $\delta^-$ end of another               | Polar molecules |

|                  |                                |                                                                                                                        |                                                                    |
|------------------|--------------------------------|------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| Hydrogen bonding | Strongest intermolecular force | A special, strong dipole–dipole interaction between a lone pair on N, O, or F and a hydrogen atom bonded to N, O, or F | NH <sub>3</sub> , H <sub>2</sub> O, HF, alcohols, carboxylic acids |
|------------------|--------------------------------|------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|

London forces increase with the number of electrons (and therefore molecular size) because larger electron clouds are more easily polarised, creating stronger temporary dipoles.

**Exam Tip:** When comparing boiling points, always identify the strongest type of intermolecular force present. Hydrogen bonding explains why water ( $M_r = 18$ ) has a much higher boiling point than methane ( $M_r = 16$ ) despite similar molecular masses.

## Giant Structures

**Giant ionic lattices** (e.g. NaCl): Strong electrostatic forces between ions in all directions → high melting points, conduct when molten/dissolved.

**Giant covalent (macromolecular) structures** (e.g. diamond, silicon dioxide, graphite): Many strong covalent bonds throughout the structure → very high melting points, generally insoluble, generally do not conduct electricity. Exception: graphite has delocalised electrons in the region between the layers, free to move *along* them, so it conducts electricity parallel to the layers (it is a poor conductor perpendicular to them); the weak London forces between layers make it soft and slippery.

**Simple molecular structures** (e.g. I<sub>2</sub>, H<sub>2</sub>O, CO<sub>2</sub>): Strong covalent bonds within molecules but weak intermolecular forces between molecules → low melting/boiling points, do not conduct electricity (no charged particles free to move).

**Exam Tip:** When explaining high melting points, always state the type of bonding/force that must be overcome and describe it as strong. For simple molecular substances, make clear it is the intermolecular forces (not the covalent bonds) that break during melting and boiling. Never say "bonds are broken" when discussing boiling — use "intermolecular forces are overcome."

## 4. Nuclear Fusion and the Origin of the Elements

The Salters storyline opens with the question of where the elements came from. The Big Bang produced essentially only hydrogen and helium (with a trace of lithium); every heavier nucleus in your body was assembled inside a star.

**Nuclear fusion:** A reaction in which two lighter nuclei join together to give a heavier nucleus.

Because all nuclei are positively charged, they repel one another strongly, so fusion only happens at the very high temperatures and pressures found in stellar cores — the nuclei need enormous kinetic energy to get close enough for the short-range strong nuclear force to take over.

Fusion is a *nuclear* change, not a chemical one. The identity of the element changes because the number of protons changes, and the energy released per gram is millions of times larger than for any chemical reaction.

## Writing Nuclear Equations

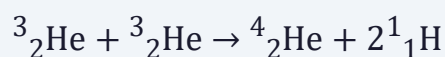
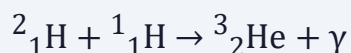
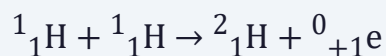
Each species is written as  ${}^A_ZX$ , where A is the mass number (protons + neutrons) and Z is the atomic number (protons). A nuclear equation balances when:

- the sum of the mass numbers is the same on both sides, and
- the sum of the atomic numbers (charges) is the same on both sides.

| Particle                        | Symbol              | Mass number | Atomic number |
|---------------------------------|---------------------|-------------|---------------|
| Proton (hydrogen nucleus)       | ${}^1_1\text{H}$    | 1           | +1            |
| Neutron                         | ${}^1_0\text{n}$    | 1           | 0             |
| Alpha particle (helium nucleus) | ${}^4_2\text{He}$   | 4           | +2            |
| Positron                        | ${}^0_{+1}\text{e}$ | 0           | +1            |
| Gamma photon                    | $\gamma$            | 0           | 0             |

## Hydrogen Burning

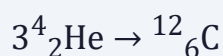
In a main-sequence star such as the Sun, four hydrogen nuclei are converted to one helium-4 nucleus through a sequence of steps:



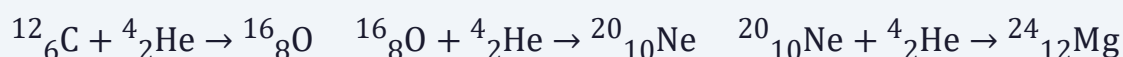
The overall change is  $4{}^1_1\text{H} \rightarrow {}^4_2\text{He} + 2{}^0_{+1}\text{e}$ . Mass numbers:  $4 = 4$ . Atomic numbers:  $4 = 2 + 2(+1) = 4$ .

## Helium Burning and Beyond

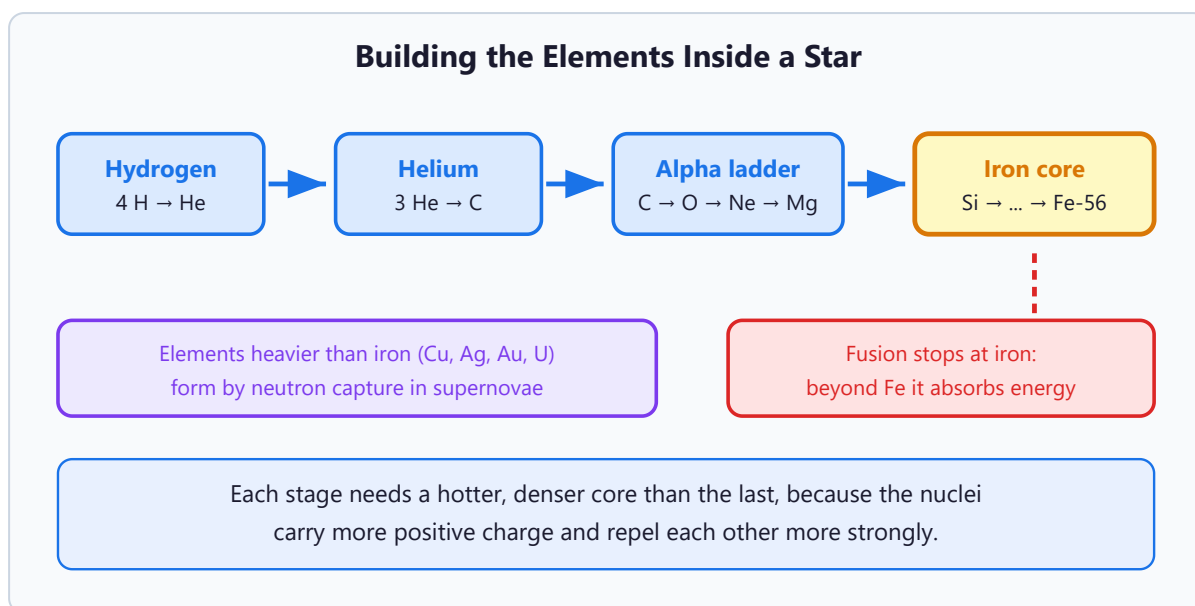
When the hydrogen in the core is exhausted, the star contracts, the core temperature rises and helium nuclei begin to fuse. The **triple-alpha process** builds carbon:



Successive additions of alpha particles then build the familiar "alpha ladder" of abundant even-numbered elements:



This explains why carbon, oxygen, neon, magnesium, silicon, sulfur and calcium — all multiples of four mass units — are far more abundant in the universe than their odd-numbered neighbours.



## Why Fusion Stops at Iron

Binding energy per nucleon rises from hydrogen up to a maximum around iron and nickel, then falls again. Fusing nuclei *lighter* than iron therefore releases energy, but fusing nuclei *heavier* than iron would absorb it. Once a massive star has built an iron-56 core it can no longer generate energy by fusion, the core collapses and the star explodes as a supernova. Elements beyond iron — copper, silver, gold, uranium — are made by nuclei capturing neutrons in and around such explosions, not by ordinary fusion.

### Worked Example — Balancing a Nuclear Equation:

Complete:  $^{14}_7\text{N} + ^4_2\text{He} \rightarrow ^{17}_8\text{O} + ?$

Mass numbers:  $14 + 4 = 18$  on the left; 17 on the right so far, so the missing particle has  $A = 18 - 17 = 1$ .

Atomic numbers:  $7 + 2 = 9$  on the left; 8 on the right so far, so the missing particle has  $Z = 9 - 8 = 1$ .

$A = 1$  and  $Z = 1$  is a proton,  $^1_1\text{H}$ . The completed equation is  $^{14}_7\text{N} + ^4_2\text{He} \rightarrow ^{17}_8\text{O} + ^1_1\text{H}$ .

**Exam Tip:** Fusion questions are almost always "balance the nuclear equation" questions, and they are free marks if you are systematic: total the top numbers, total the bottom numbers, and read off what is missing. Do not confuse fusion (light nuclei joining, needs very high temperature and pressure)

with fission (a heavy nucleus splitting). And note that a positron has mass number 0 but atomic number +1 — forgetting it is the commonest way to get the charge balance wrong.

## 5. Group 2 Chemistry

Group 2 — the alkaline earth metals — enters the storyline through the salts found in sea water, in bath salts, in barium meals and in fertilisers. H433 requires magnesium, calcium, strontium and barium.

### Trends Down the Group

Each element has the outer configuration  $ns^2$  and forms a  $2+$  ion by losing both outer electrons. Going down the group:

- atomic radius increases (an extra occupied shell each time);
- shielding by inner shells increases;
- so the outer electrons are attracted less strongly and the first (and second) ionisation enthalpies **decrease**;
- so the elements lose electrons more easily and **reactivity increases** down the group.

**Exam Tip:** Nuclear charge does increase down the group, so you must say why it loses: the increase in distance and shielding outweighs the increase in nuclear charge. An answer that only says "more shells" or only says "more protons" will not get full marks.

### Reaction with Water

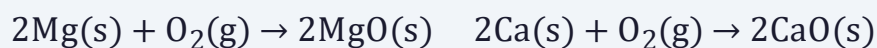
| Element | With cold water                                                                        | Equation                                                                   |
|---------|----------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Mg      | Extremely slow; reacts readily with steam instead, burning with a bright white flame   | $\text{Mg} + \text{H}_2\text{O(g)} \rightarrow \text{MgO} + \text{H}_2$    |
| Ca      | Steady effervescence; a cloudy white suspension of sparingly soluble $\text{Ca(OH)}_2$ | $\text{Ca} + 2\text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2 + \text{H}_2$ |

|    |                                                                              |                                                                                   |
|----|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Sr | Faster effervescence                                                         | $\text{Sr} + 2\text{H}_2\text{O} \rightarrow \text{Sr}(\text{OH})_2 + \text{H}_2$ |
| Ba | Vigorous effervescence; the hydroxide is soluble so the solution stays clear | $\text{Ba} + 2\text{H}_2\text{O} \rightarrow \text{Ba}(\text{OH})_2 + \text{H}_2$ |

In every case the metal is oxidised from 0 to +2 and hydrogen in water is reduced from +1 to 0, so these are redox reactions. The solutions formed are alkaline — hence "alkaline earth" metals.

## Reaction with Oxygen

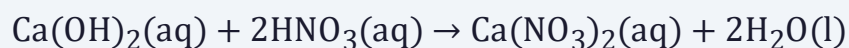
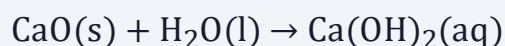
All four burn in oxygen to give the white ionic oxide:



Magnesium burns with the familiar brilliant white flame. Reactivity towards oxygen also increases down the group, and barium is stored under oil because it tarnishes so quickly in air.

## Oxides and Hydroxides are Basic

Group 2 oxides react with water to give the hydroxide, and both oxides and hydroxides neutralise acids:



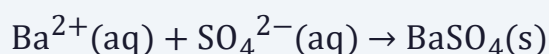
Because hydroxide solubility rises down the group, the pH of the saturated solution also rises: a saturated  $\text{Mg}(\text{OH})_2$  solution is only weakly alkaline (about pH 10), while saturated  $\text{Ba}(\text{OH})_2$  is strongly alkaline.

## Solubility Trends

| Compound type | Trend down the group | Illustration |
|---------------|----------------------|--------------|
|---------------|----------------------|--------------|

|                       |                             |                                                                                  |
|-----------------------|-----------------------------|----------------------------------------------------------------------------------|
| Hydroxides, $M(OH)_2$ | Solubility <b>increases</b> | $Mg(OH)_2$ sparingly soluble (milk of magnesia) $\rightarrow$ $Ba(OH)_2$ soluble |
| Sulfates, $MSO_4$     | Solubility <b>decreases</b> | $MgSO_4$ soluble (Epsom salts) $\rightarrow$ $BaSO_4$ essentially insoluble      |
| Carbonates, $MCO_3$   | All insoluble               | $CaCO_3$ is limestone, chalk and seashell                                        |

The two opposite trends are the basis of two standard tests. Barium sulfate is so insoluble that adding acidified barium chloride solution to an unknown gives a **white precipitate** only if sulfate ions are present:



It is also why a "barium meal" is safe to swallow for an X-ray: soluble barium salts are highly toxic, but barium sulfate does not dissolve in the gut, so essentially no  $Ba^{2+}(aq)$  is absorbed. The acid is added first to destroy any carbonate, which would otherwise also give a white precipitate.

## Thermal Stability of the Carbonates

All Group 2 carbonates decompose on strong heating to the oxide and carbon dioxide:



| Carbonate | Ionic radius of $M^{2+}$ / nm | Approximate decomposition temperature / °C |
|-----------|-------------------------------|--------------------------------------------|
| $MgCO_3$  | 0.072                         | 540                                        |
| $CaCO_3$  | 0.100                         | 900                                        |
| $SrCO_3$  | 0.118                         | 1290                                       |
| $BaCO_3$  | 0.135                         | 1360                                       |

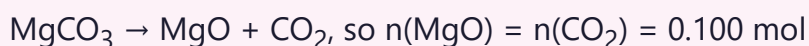
**Charge density:** The ratio of an ion's charge to its size. For a given charge, a smaller ion has a higher charge density and therefore a more concentrated electric field around it.

The explanation runs through polarisation.  $\text{Mg}^{2+}$  is small and 2+, so it has a high charge density. It attracts electron density from the large, easily distorted carbonate ion, weakening one of the C–O bonds and making it easier to break away as  $\text{CO}_2$ . Going down the group the cation gets larger, its charge density falls, the carbonate ion is distorted less, and a higher temperature is needed. So **thermal stability increases down the group**. The same argument explains why Group 1 carbonates (1+ ions, lower charge density) are far more stable than Group 2 carbonates of similar size.

### Worked Example — Decomposing a Carbonate:

8.43 g of magnesium carbonate is heated strongly until there is no further mass change. Calculate the mass of solid remaining and the volume of carbon dioxide produced at RTP. ( $M_r$ :  $\text{MgCO}_3 = 84.3$ ,  $\text{MgO} = 40.3$ ; molar gas volume =  $24.0 \text{ dm}^3 \text{ mol}^{-1}$ )

$$n(\text{MgCO}_3) = 8.43 / 84.3 = 0.100 \text{ mol}$$



$$\text{Mass of MgO} = 0.100 \times 40.3 = \mathbf{4.03 \text{ g}}$$

$$\text{Volume of CO}_2 = 0.100 \times 24.0 = \mathbf{2.40 \text{ dm}^3}$$

Check: mass lost =  $8.43 - 4.03 = 4.40 \text{ g}$ , which is 0.100 mol of  $\text{CO}_2$  ( $M_r = 44.0$ ). Consistent.

## Where Group 2 Compounds Are Used

| Compound                 | Use                                           | Why that compound                                                                                      |
|--------------------------|-----------------------------------------------|--------------------------------------------------------------------------------------------------------|
| $\text{Mg}(\text{OH})_2$ | Antacid ("milk of magnesia")                  | A base, but only sparingly soluble, so it neutralises stomach acid without making the stomach alkaline |
| $\text{CaCO}_3$          | Indigestion tablets; flue-gas desulfurisation | Cheap, mild base that neutralises acid                                                                 |
| $\text{Ca}(\text{OH})_2$ | Agricultural lime, spread on acidic soil      | Soluble enough to raise soil pH quickly                                                                |

|                                      |                                            |                                                                                    |
|--------------------------------------|--------------------------------------------|------------------------------------------------------------------------------------|
| BaSO <sub>4</sub>                    | "Barium meal" for X-ray imaging of the gut | Opaque to X-rays and so insoluble that toxic Ba <sup>2+</sup> (aq) is not absorbed |
| MgSO <sub>4</sub> ·7H <sub>2</sub> O | Epsom salts, used in bath salts            | Very soluble, so it dissolves readily in bath water                                |

**Exam Tip:** Two trends in this section run in opposite directions and are easy to mix up. Hydroxide solubility *increases* down the group; sulfate solubility *decreases*. A useful anchor is the pair of practical facts: Ca(OH)<sub>2</sub> is used as limewater because it dissolves a little, and BaSO<sub>4</sub> is safe to swallow because it does not dissolve at all.

## 6. Atomic Spectra and Electron Energy Levels

The storyline reaches this point by asking a simple question: how can we possibly know what a star is made of? The answer is that the light arriving from it carries a fingerprint of the elements in it, and reading that fingerprint is the strongest single piece of evidence we have that electrons in atoms are confined to fixed energy levels.

### The Electromagnetic Spectrum

Visible light is one narrow band of a much wider spectrum of electromagnetic radiation. The three regions H433 requires you to place in order are infrared, visible and ultraviolet:

| Region      | Frequency | Wavelength | Energy per photon | What it does to a molecule                                             |
|-------------|-----------|------------|-------------------|------------------------------------------------------------------------|
| Infrared    | Lowest    | Longest    | Lowest            | Makes bonds vibrate — the basis of IR spectroscopy in WM               |
| Visible     | Middle    | Middle     | Middle            | Promotes electrons in delocalised systems — the origin of colour in CD |
| Ultraviolet | Highest   | Shortest   | Highest           | Breaks bonds to form radicals — photodissociation in OZ                |

Frequency, wavelength and energy are tied together by two relationships that appear on the H433 Data Sheet:

$$E = h\nu \quad c = \nu\lambda$$

Here  $E$  is the energy of one photon in J,  $h$  is the Planck constant ( $6.63 \times 10^{-34}$  J s),  $\nu$  is the frequency in Hz ( $\text{s}^{-1}$ ),  $\lambda$  is the wavelength in m and  $c$  is the speed of electromagnetic radiation,  $3.00 \times 10^8$  m  $\text{s}^{-1}$ . Combining them gives  $E = hc/\lambda$ , so high frequency means short wavelength means high energy.

## Emission and Absorption Spectra

An electron in an atom can only have certain allowed energies. It cannot sit between levels, so it can only move from one allowed level to another — and when it does, the energy difference must be carried away or supplied as a single photon of exactly that energy:

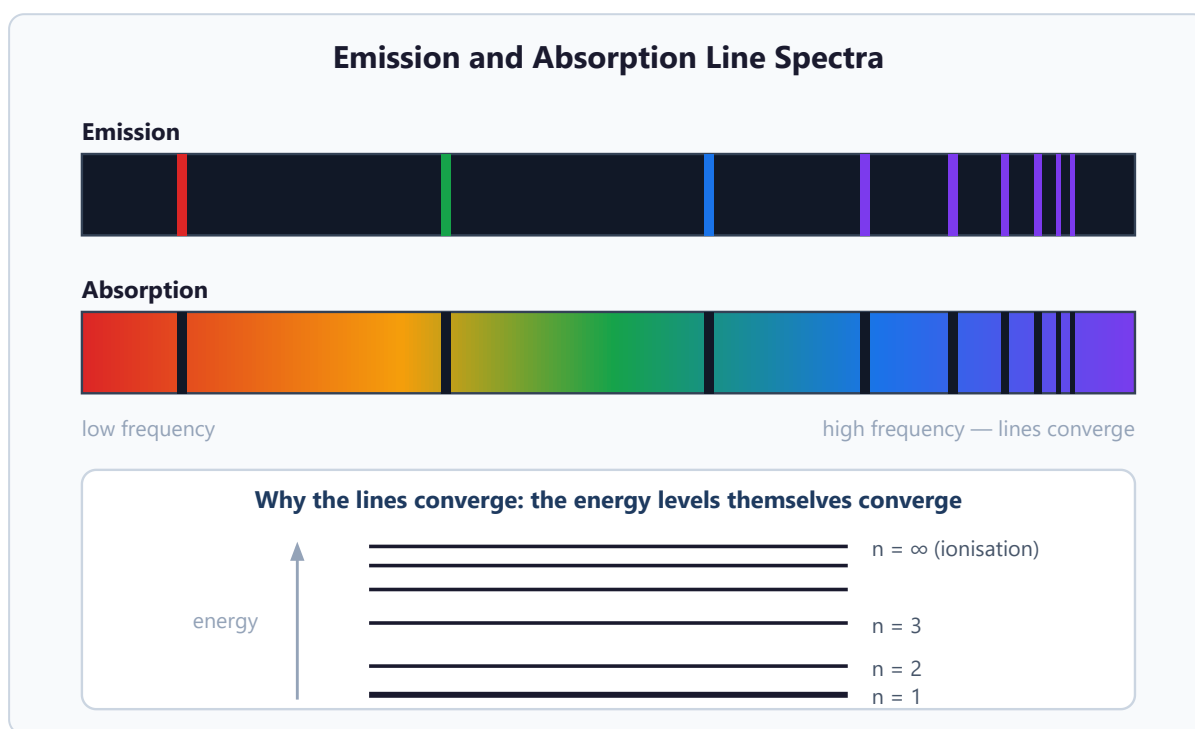
$$\Delta E = h\nu$$

That single sentence explains both kinds of spectrum.

- In an **emission spectrum**, atoms are given energy (heated in a flame, or excited by an electric discharge in a gas). Electrons jump up to higher levels, then fall back down, each drop emitting a photon of one particular frequency. The result is a set of bright coloured lines on a black background.
- In an **absorption spectrum**, white light containing all frequencies is passed through a cool sample of the element. Photons whose energy exactly matches a gap between levels are absorbed to lift electrons up; every other frequency passes through. The result is a set of black lines on a continuous coloured background.

|                                | Emission spectrum                                                                        | Absorption spectrum                                     |
|--------------------------------|------------------------------------------------------------------------------------------|---------------------------------------------------------|
| Appearance                     | Bright coloured lines on a black background                                              | Black lines on a bright, continuous coloured background |
| Direction of the electron jump | Down, from a higher to a lower level                                                     | Up, from a lower to a higher level                      |
| Line positions                 | Identical for a given element — the lines appear at exactly the same frequencies in both |                                                         |

|                  |                                                                                             |
|------------------|---------------------------------------------------------------------------------------------|
| Type of spectrum | Both are <b>line</b> spectra, not continuous                                                |
| Spacing of lines | Lines converge (get closer together) towards higher frequency                               |
| Structure        | Both contain series of lines, each series being transitions to or from one particular level |



## The Convergence Limit and Ionisation

The energy levels get closer together as they get further from the nucleus, so the jumps between them get smaller and the lines in each series crowd together at the high-frequency end until they merge. That point is the **convergence limit**, and it corresponds to the electron being removed from the atom altogether — in other words, to ionisation. Measuring the frequency at the convergence limit therefore gives a direct route to the ionisation enthalpy.

### Worked Example — Ionisation Enthalpy from a Spectrum:

The series of lines in the hydrogen emission spectrum that end at the  $n = 1$  level converge at a frequency of  $3.28 \times 10^{15}$  Hz. Calculate the first ionisation enthalpy of hydrogen in  $\text{kJ mol}^{-1}$ .

( $h = 6.63 \times 10^{-34} \text{ J s}$ ,  $L = 6.02 \times 10^{23} \text{ mol}^{-1}$ )

Energy for one atom:  $\Delta E = h\nu = 6.63 \times 10^{-34} \times 3.28 \times 10^{15} = 2.175 \times 10^{-18} \text{ J}$

Energy for one mole:  $2.175 \times 10^{-18} \times 6.02 \times 10^{23} = 1.309 \times 10^6 \text{ J mol}^{-1}$

First ionisation enthalpy = **1310 kJ mol<sup>-1</sup>** (3 s.f.), which agrees with the measured value of 1312 kJ mol<sup>-1</sup>.

### Worked Example — Energy of a Sodium Photon:

The yellow line in the sodium emission spectrum has a wavelength of 589 nm. Calculate its frequency and the energy of one photon.

$$\lambda = 589 \text{ nm} = 589 \times 10^{-9} \text{ m} = 5.89 \times 10^{-7} \text{ m}$$

$$\nu = c/\lambda = 3.00 \times 10^8 / 5.89 \times 10^{-7} = \mathbf{5.09 \times 10^{14} \text{ Hz}}$$

$$E = h\nu = 6.63 \times 10^{-34} \times 5.09 \times 10^{14} = \mathbf{3.37 \times 10^{-19} \text{ J}}$$

This is the transition responsible for the yellow colour of a sodium street lamp and of a sodium flame test.

## Flame Colours

A flame test is an emission spectrum you can see with the naked eye. Heat gives the outer electrons of the metal ion enough energy to jump to a higher level; as they drop back, they emit visible photons whose frequencies depend on the spacing of that ion's energy levels, and since every element has its own spacing, every element has its own colour. Clean a nichrome or platinum wire with concentrated hydrochloric acid, dip it in the solid, and hold it in the hot blue part of a Bunsen flame.

| Ion             | Flame colour    |
|-----------------|-----------------|
| Li <sup>+</sup> | Red (crimson)   |
| Na <sup>+</sup> | Yellow / orange |
| K <sup>+</sup>  | Lilac           |

|                  |                          |
|------------------|--------------------------|
| $\text{Ca}^{2+}$ | Orange-red (brick red)   |
| $\text{Ba}^{2+}$ | Pale green (apple green) |
| $\text{Cu}^{2+}$ | Blue-green               |

**Exam Tip:** The commonest error in calculations here is unit conversion:  $\lambda$  must be in metres before you use  $c = v\lambda$ , so 589 nm is  $5.89 \times 10^{-7}$  m, not  $5.89 \times 10^{-9}$  m. The second commonest is forgetting to multiply by the Avogadro constant when the answer is wanted per mole, and the third is leaving the answer in J when  $\text{kJ mol}^{-1}$  is asked for. In explanation questions, always say that the spectrum is a *line* spectrum rather than a continuous one, because that is the point: discrete lines can only arise if the electron energies are themselves discrete.

## 7. Identifying Ions: Solubility, Precipitates and Tests

Blood, sea water and the salts used in medicine and agriculture are all solutions of ions, and the practical question the storyline poses is how you find out which ions a sample contains. The method rests entirely on solubility: if you add a reagent and a solid appears, the two ions you have just brought together form an insoluble compound.

**Precipitation reaction:** A reaction in which two soluble ionic compounds are mixed in solution and a new, insoluble compound forms as a solid. Only the two ions that combine appear in the ionic equation; the others are spectators.

### Solubility Rules

Learn these as a short list of "all ... except ..." statements rather than compound by compound.

| Anion           | Soluble                         | Insoluble |
|-----------------|---------------------------------|-----------|
| $\text{NO}_3^-$ | All nitrates, without exception | —         |

|                                              |                                                                                                  |                                                                            |
|----------------------------------------------|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| $\text{Cl}^-$ , $\text{Br}^-$ , $\text{I}^-$ | Most halides                                                                                     | $\text{Ag}^+$ and $\text{Pb}^{2+}$ halides                                 |
| $\text{SO}_4^{2-}$                           | Most sulfates                                                                                    | $\text{BaSO}_4$ , $\text{PbSO}_4$ ( $\text{CaSO}_4$ only slightly soluble) |
| $\text{CO}_3^{2-}$                           | Only $\text{Li}^+$ , $\text{Na}^+$ , $\text{K}^+$ and $\text{NH}_4^+$ carbonates                 | All the rest                                                               |
| $\text{OH}^-$                                | Only $\text{Na}^+$ , $\text{K}^+$ and $\text{Ba}^{2+}$ hydroxides ( $\text{Ca(OH)}_2$ sparingly) | All the rest                                                               |

On the cation side the rule is even shorter: **every** compound of  $\text{Li}^+$ ,  $\text{Na}^+$ ,  $\text{K}^+$  and  $\text{NH}_4^+$  is soluble, so those ions can never be precipitated and must be identified another way — by flame test for the first three, and by warming with sodium hydroxide for ammonium.

## Precipitate Colours

The colour of the solid, and what happens when you add more reagent, is what actually identifies the ion.

| Reagent added            | Ion present      | Precipitate and colour                                                              | With excess reagent    |
|--------------------------|------------------|-------------------------------------------------------------------------------------|------------------------|
| Dilute $\text{NaOH(aq)}$ | $\text{Cu}^{2+}$ | $\text{Cu(OH)}_2$ , pale blue                                                       | Insoluble              |
|                          | $\text{Fe}^{2+}$ | $\text{Fe(OH)}_2$ , dirty green — turns red-brown at the surface on standing in air | Insoluble              |
|                          | $\text{Fe}^{3+}$ | $\text{Fe(OH)}_3$ , red-brown (rust)                                                | Insoluble              |
|                          | $\text{Al}^{3+}$ | $\text{Al(OH)}_3$ , white                                                           | Dissolves — amphoteric |
|                          | $\text{Zn}^{2+}$ | $\text{Zn(OH)}_2$ , white                                                           | Dissolves — amphoteric |
|                          | $\text{NH}_4^+$  | No precipitate; on warming, a pungent gas that turns damp red litmus blue           | —                      |

|                                                                       |                               |                                                         |                                               |
|-----------------------------------------------------------------------|-------------------------------|---------------------------------------------------------|-----------------------------------------------|
| AgNO <sub>3</sub> (aq), after acidifying with dilute HNO <sub>3</sub> | Cl <sup>-</sup>               | AgCl, white                                             | Dissolves in <i>dilute</i> ammonia            |
|                                                                       | Br <sup>-</sup>               | AgBr, cream                                             | Dissolves only in <i>concentrated</i> ammonia |
|                                                                       | I <sup>-</sup>                | AgI, yellow                                             | Insoluble even in concentrated ammonia        |
| BaCl <sub>2</sub> (aq), after acidifying with dilute HCl              | SO <sub>4</sub> <sup>2-</sup> | BaSO <sub>4</sub> , white                               | Insoluble                                     |
| Dilute HCl or HNO <sub>3</sub>                                        | CO <sub>3</sub> <sup>2-</sup> | No solid — effervescence; the gas turns limewater milky | —                                             |

The colour of the solution itself is a clue before any reagent is added: Cu<sup>2+</sup>(aq) is blue, Fe<sup>2+</sup>(aq) pale green and Fe<sup>3+</sup>(aq) yellow-brown, while Na<sup>+</sup>, K<sup>+</sup>, Ca<sup>2+</sup>, Ba<sup>2+</sup>, Zn<sup>2+</sup>, Al<sup>3+</sup> and Pb<sup>2+</sup> solutions are all colourless.

## A Sequence of Tests to Identify a Salt

Given an unknown salt containing one cation and one anion from the lists above, the order of the tests matters, because an early test can ruin a later one. Work through them like this:

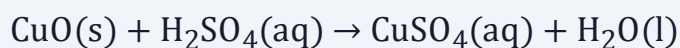
- Look at it.** Note the colour of the solid and of its solution in water. If it dissolves at all, the cation is unlikely to be one that forms only insoluble compounds.
- Flame test.** Clean the wire with concentrated HCl, then test: red for Li<sup>+</sup>, yellow for Na<sup>+</sup>, lilac for K<sup>+</sup>, orange-red for Ca<sup>2+</sup>, pale green for Ba<sup>2+</sup>, blue-green for Cu<sup>2+</sup>.
- Add dilute NaOH(aq), then excess.** Match the precipitate colour against the table. Warm the mixture and test any gas with damp red litmus to check for NH<sub>4</sub><sup>+</sup>.
- Add dilute acid.** Effervescence, with a gas that turns limewater milky, confirms carbonate. This must come *before* the next two tests, because carbonate would give a white precipitate with both silver and barium ions and be mistaken for chloride or sulfate.
- Acidify with dilute nitric acid, then add silver nitrate.** White, cream or yellow identifies Cl<sup>-</sup>, Br<sup>-</sup> or I<sup>-</sup>; confirm by testing solubility of the precipitate in dilute and then concentrated ammonia.

6. **Acidify a fresh portion with dilute hydrochloric acid, then add barium chloride.** A white precipitate confirms  $\text{SO}_4^{2-}$ .
7. **If no anion test is positive**, and the salt is soluble, the anion is nitrate — every nitrate is soluble and gives no precipitate with any of these reagents.

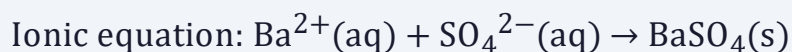
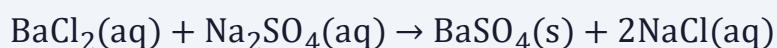
## Making Salts

The same solubility rules tell you which of the two standard preparative routes to use.

- **A soluble salt** is made by reacting a dilute acid with an excess of an insoluble base — a metal oxide, hydroxide or carbonate — warming to complete the reaction, filtering off the unreacted excess, and then crystallising the filtrate by evaporating it to the point of crystallisation and leaving it to cool. Using an excess of the solid base guarantees that no acid is left in the product.



- **An insoluble salt** is made by precipitation: mix solutions of two soluble salts chosen so that the ions you want come together, then filter, wash the residue with distilled water to remove the soluble spectator ions, and dry it.



**Exam Tip:** Two details are worth a mark each and are routinely dropped. Always say *why* the solution is acidified before adding silver nitrate or barium chloride — to remove carbonate ions, which would otherwise give a white precipitate and a false positive. And always use the matching acid: nitric acid before silver nitrate (hydrochloric acid would add chloride ions and give an instant white precipitate whatever the sample), and hydrochloric acid before barium chloride (sulfuric acid would add the very ion you are testing for).